



Ify Bidokwu (Attempt 1 of 1)

Test Information

QUESTION 1: CALCULATED NUMERIC

5

out of 5 points

A controller sends an 12mA analog signal to an I/P converter. The pressure signal from the I/P converter is sent to the control valve. What is the valve opening as a percentage of the total travel, assuming total travel corresponds to 100% (fully open valve)



Given Answer: 50

Correct Answer: 50

Answer range +/- 0 (50 - 50)

QUESTION 2: MULTIPLE ANSWER

10

out of 10 points

Consider the process transfer function: $G_P(s) = \frac{1}{-5s+1}$ Assume that the controller gain K_c is positive. Using the Routh–Hurwitz criterion, determine which the following controllers can stabilize the loop. Select all that apply.

Given Proportional Derivative (PD): $G_c(s) = K_c(1 + \tau_D s)$

Answers:

Proportional Integral Derivative (PID): $G_c(s) = K_c\left(1 + \frac{1}{\tau_I s} + \tau_D s\right)$

Correct Proportional Derivative (PD): $G_c(s) = K_c(1 + \tau_D s)$

Answers:

Proportional Integral Derivative (PID): $G_c(s) = K_c\left(1 + \frac{1}{\tau_I s} + \tau_D s\right)$

QUESTION 3: SHORT ANSWER

10

out of 10 points

Consider the process transfer function: $G_P(s) = \frac{1}{-5s+1}$ Assume that the controller gain K_c is positive. Select a controller that stabilizes the closed loop response and derive the stability conditions on the controller parameters. Write your answer in the textbox below.

Given Answer: The process by itself is unstable. The derivative term must be big enough to overcome the 5 second benchmark in the denominator and make the time constant of the closed loop negative or stable.

$$\tau_D > 5/K_c$$

Correct [None]
Answer:

Response For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

[illegible]

QUESTION 4: SHORT ANSWER

10

out of 10 points

Consider the mixing of two streams: **Stream 1** with flow rate $F_1=5$ kg/s and temperature $T_1=25$ °C and **Stream 2** with flow rate $F_2=5$ kg/s and temperature $T_2=75$ °C. The streams are mixed in a tank with constant mass (from a perfect level controller) equal to $M=100$ kg. We wish to design a controller that would regulate the temperature of the effluent stream from the tank by manipulating the flow rate of **Stream 1**. Derive the process transfer function for this control loop. Your answer must contain only numerical values and the Laplace transformation symbol s . Temperature units in the final answer must be in °C. Assume that the heat capacities of all streams and that of the mass in the mixing tank are all equal.

Given $\frac{-2.5}{10s + 1}$
 Answer:

Correct [None]
Answer:

Response For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).



QUESTION 5: CALCULATED NUMERIC

0

out of 5 points

Consider a process described by the following transfer function

$$G_P(s) = \frac{(-s+1)}{(10s+1)(2s+1)}$$



We wish to approximate this process with an FOPTD. Calculate the Delay time of the approximate model.

Given Answer: 3

Correct Answer: 2

Answer range +/- 0 (2 - 2)

QUESTION 6: CALCULATED NUMERIC

0

out of 5 points

Determine the Open Stem Travel (%Valve Open) of a 4in inline control valve in an 8in NPS Schedule 40 pipeline, with inherent characteristic given in **Table 1** in the **CHE Calculator** that is used to regulate the flow of liquid propane. The service conditions are: Flowing temperature $T=50^{\circ}\text{F}$, volumetric design flow rate $Q=250$ gpm, upstream pressure $P_1=350$ psia and downstream pressure $P_2=300$ psia. Express your result in percent travel (for instance if you



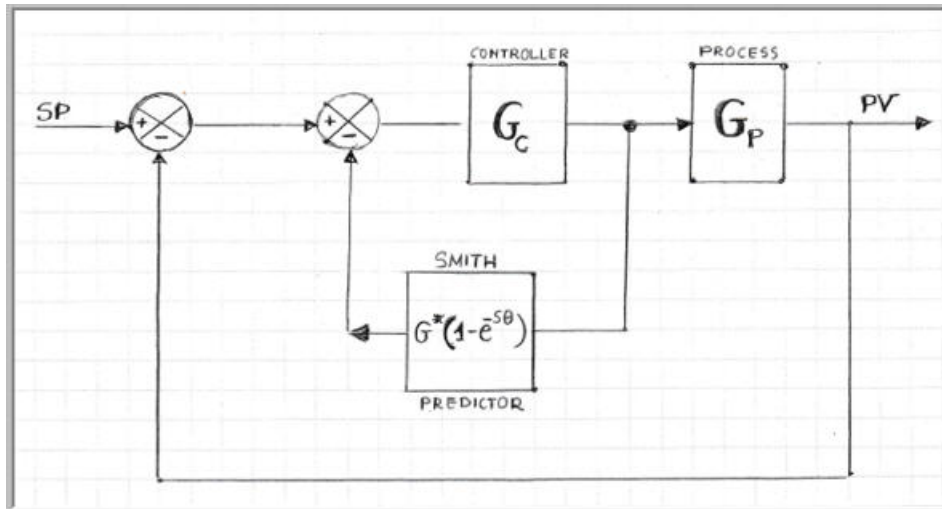
answer is 25.3% enter the value 25). Assume that the **Peng-Robinson** Equation of State can be used for the calculation of the thermal properties of propane and the **Rackett** equation for the liquid density. **Round your answer to a whole number.**

Given Answer: [None Given]

Correct Answer: 45

Answer range +/- 0 (45 - 45)

Time delays frequently occur in the process industries because of the presence of transportation lags, recycle loops and sensor inertia. Time delays result in reduction of the controller gain in order to maintain stability and as a result make the controller performance sluggish. To improve the performance of loops with time delays, advanced control strategies have been developed over the years. One such strategy is known as the **Smith predictor** and it consists of an ordinary feedback loop plus an inner loop that introduces an extra term directly into the feedback path. A control block diagram of the Smith predictor controller is given below:



where $G_P(s) = G_P^*(s) e^{-s\theta}$ and $G_P^*(s)$ is the process transfer function free of the time delay. Consider the



following first order process $G_P(s) = \frac{1.3e^{-3.7s}}{13.7s+1}$. This chemical process is controlled with a PI feedback

controller **with the Smith-Predictor in the loop**. The delay time in the Smith-Predictor is set at $\theta = 3.7$. Perform a simulation for a unit step change in the set point and calculate the **overshoot** and the **settling time**. Enter your numerical results separated by comma in the textbox below:

Given: Overshoot = 0
 Answer: Settling time = 18.85
 (0,18.85)

Correct: [None]
 Answer:

Response Feedback: For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 8: CALCULATED NUMERIC

5

out of 5 points

Consider the control of a process with transfer function

$$G_P(s) = \frac{0.5e^{-5s}}{(20s+1)(5s+1)}$$



For a proportional controller with gain K_C at what value of the gain the process begins to become unstable. Round your numerical answer into **four decimal digits**.

Given Answer: 11.9597

Correct Answer: 11.9598

Answer range +/- 0.0001 (11.95970 - 11.95990)

QUESTION 9: SHORT ANSWER

0

out of 10 points

Consider a first order process with time delay $G_P(s) = \frac{1.2e^{-3.7s}}{13.7s+1}$. The process is regulated with a PI controller with Gain $K_C = 0.45 K_u$ and Reset Time $\tau_I = 0.83 P_u$. Perform a simulation for a unit size step change in the **Set Point** and calculate the overshoot and the settling time. Enter

1. The additional time delay the loop can tolerate before it becomes unstable
2. The overshoot to the step change in the set point
3. The settling time after the step change in the set point

Enter all values separated by commas, in the textbox below:

Given Answer: 2. Overshoot = 0.0158

3. Settling Time = 14.13

Correct Answer: [None]

Answer:



Response Feedback: For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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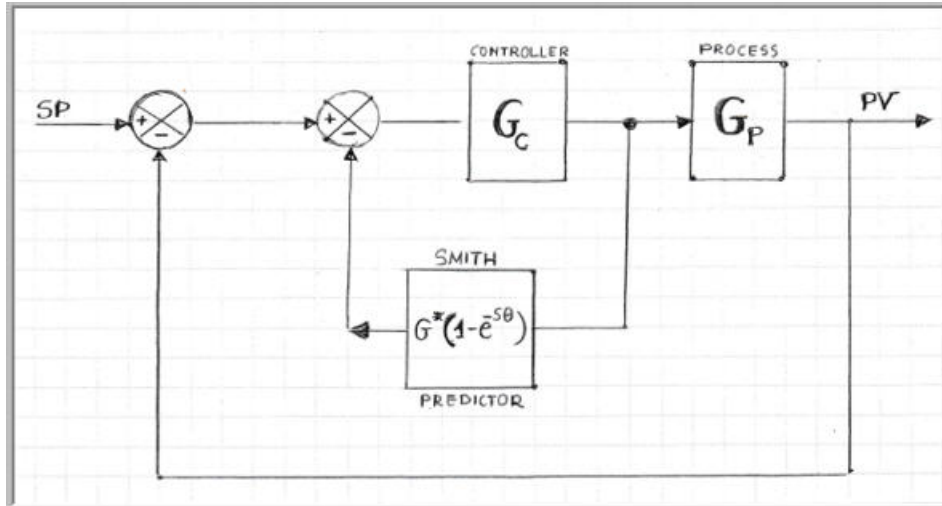
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Time delays frequently occur in the process industries because of the presence of transportation lags, recycle loops and sensor inertia. Time delays result in reduction of the controller gain in order to maintain stability and as a result make the controller performance sluggish. To improve the performance of loops with time delays, advanced control strategies have been developed over the years. One such strategy is known as the **Smith predictor** and it consists of an ordinary feedback loop plus an inner loop that introduces an extra term directly into the feedback path. A control block diagram of the Smith predictor controller is given below:



where $G_P(s) = G_P^*(s) e^{-s\theta}$ and $G_P^*(s)$ is the process transfer function free of the time delay. Derive the



closed-loop transfer function $G_{CL}(s) = \frac{PV}{SP}$ and enter your final formula in the textbox below:

Given
Answer:
$$\frac{G_C G_P^* e^{-s\theta}}{1 + G_C G_P^* e^{-s\theta}}$$

Correct
Answer: [None]

Response Feedback: For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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QUESTION 11: CALCULATED NUMERIC

5

out of 5 points

Using the Bode stability analysis determine the minimum value of the reset time τ_I for stability of a **PI** controller with gain $K_C = 0.3$ applied to a process with transfer function

$$G_P(s) = \frac{e^{-0.5s}}{s+1}$$



Round your numerical answer into **four decimal digits**.

Given Answer: ✓ 0.1117

Correct Answer: ✓ 0.1118

Answer range +/- 0.0001 (0.11170 - 0.11190)

FEEDBACK AND NOTES FOR ATTEMPT